

34 — Advisory warnings on an otherwise valid input

Proves · Advisory warnings only — results are still computed and shown.

Mechanics this scenario turns on

- Validation runs **before** Level 1. A plant that cannot carry its load is rejected there, with a different message and a different code path from the Level 1 rejections.

Panels described below · The two triggers · The response · Why both warnings carry `type: "battery"`

Anything not described here — the mechanics above name the step that produced it; `00-ORIENTATION` Part 6 has the full number-to-step index.

Trust · characterisation snapshot, generated from the code. It detects change; it does NOT prove correctness. Figures marked *pending reference verification* have never been checked against anything outside the application.

Read after · scenario 17.

Warnings come in two severities and they behave completely differently:

```
Severity.Error    → promoted into ValidationResult.Errors → the request 400s
Severity.Warning  → advisory only → the request succeeds, the client shows a banner
```

Before this scenario only **two** advisory warnings had golden coverage (battery capacity, scenario 10; operating hours, scenario 14). This adds the remaining two, on one otherwise perfectly valid input.

The two triggers

```
"battery": { "powerKw": 500, "capacityKwh": 1000, "relevantModes": [] },
"dpRedundancyRequirementKw": 400           // with dpEnabled absent
```

Battery without modes. `BatteryConfigurationInput.IsActive` requires `PowerKw > 0` **and** at least one relevant mode. With an empty list the battery is inert — no cascade, no panel, no effect on any number. A user who typed 500 kW and expected something to change gets told why nothing did.

DP redundancy without DP. The redundancy figure only participates in a DP-mode allocation. With DP off it is stored, exported to the saved profile, and ignored. The warning says so.

The response

```
200
baseline FOC 13 652.6 t/yr    (identical to scenario 03 – the battery really is inert)
warnings:
[Warning] battery: Battery power is configured but no relevant modes are selected ...
[Warning] battery: DP redundancy requirement is set but DP mode is not enabled ...
```

Note `batteryDetails` is **null**, not an empty panel — decision G2/B10, the same rule scenario 18 covers from the zero-hours angle.

Why both warnings carry `type: "battery"`

They are grouped for the client's banner. If a future change needs to style them differently, the `type` field is the thing to split — and this snapshot is where that change would show up.

Takeaway: the calculator answers even when part of the input is pointless, and says which part. Silence would be worse than a wrong number here.

KSailCalc manual test scenarios — calculation walkthrough · regenerated 2026-08-10 · numbers verified against commit 559aef2 and unchanged by the backend/client refactors (18 golden snapshots byte-identical)